



【發明摘要】

【中文發明名稱】運用於三維結構與應力表徵之混合式偏振數位全像顯微量測系統與方法

【英文發明名稱】 HYBRID POLARIZED DIGITAL HOLOGRAPHIC MICROSCOPY SYSTEM AND METHODS FOR 3D STRUCTURAL AND STRESS METROLOGY

【中文】

本發明提供一種用於檢測表面形貌以及應力特性之混合式偏振數位全像顯微量測系統，包括光源模組、偏振調制模組、分光元件以及偵測與分析模組。光源模組用以產生第一偵測光以及第二偵測光。偏振調制模組設置於第一偵測光以及第二偵測光的光路上，用以將第一偵測光調制成圓偏極偵測光通過待測物以形成測物光，以及將第二偵測光調制成離軸圓偏極參考光。分光元件用以接收測物光以及離軸圓偏極參考光並將其分光，再於偵測與分析模組相應的第一與第二偵測平面上干涉形成第一干涉影像以及第二干涉影像，偵測與分析模組根據第一干涉影像以及第二干涉影像建立關於待測物的三維表面形貌以及三維應力分布。

【英文】

A hybrid polarized digital holographic microscopy system and method for detecting three dimensional structural and stress characteristics, comprising a light source module, a polarization modulation module, a beam splitting element, and a detection and analysis module. The light source module is configured to generate a first detection beam and a second detection beam. The polarization modulation module is disposed on the optical paths of the first and second detection beams to modulate the first detection

beam into a circularly polarized detection beam that passes through an object under test to form an object beam, and to modulate the second detection beam into an off-axis circularly polarized reference beam. The beam splitting element is configured to receive and split the object beam and the off-axis circularly polarized reference beam, which subsequently interfere to form a first interference image and a second interference image on corresponding first and second detection planes of the detection and analysis module. Based on the first and second interference images, the detection and analysis module establishes the three-dimensional (3D) surface topography and 3D stress distribution of the object under test.